Rules First, Models Second: Detecting Personal Data in English, French, German and Luxembourgish Text

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https://github.com/YannKleees/tagging

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Abstract

Documents written in Luxembourg mix four languages and carry identifiers that general tools for detecting personal data do not know. This paper describes *tagging*, an open source tool that combines a rule engine written in Rust with two optional models for named entity recognition, and compares it with Microsoft Presidio on the same data. The rules cover formats with check digits, word lists from Wikidata, a grammar for names, labelled form fields and addresses. A list of 190,000 everyday words, counted from Wikipedia text and taken from the official Luxembourgish dictionary, removes model output that is no name. On template sentences in Luxembourg's formats with names the word lists had not been built around, the tool covered 0.93 to 0.97 of all labelled items with a model switched on, against 0.87 for Presidio. On synthetic letters and forms from another generator it covered 0.75 to 0.84 against 0.49 to 0.59. On text without personal data it marked 7 to 16 spans where Presidio marked 71 to 115. Presidio remained ahead on names in German and Luxembourgish Wikipedia sentences (0.87 and 0.83 against 0.85 and 0.77), and it was 18 times faster than the more accurate of the two models. The rules alone found 0.98 of the persons in the first test set and 0.54 once the names were replaced, which shows how much a saturated test set can hide. All data were synthetic or encyclopedic, so performance on real documents remains unmeasured.

Keywords: personal data, de-identification, named entity recognition, Luxembourgish, rule-based systems, evaluation

1 Introduction

The General Data Protection Regulation asks organisations to limit the personal data they keep and share [1]. In practice this often means removing names, addresses and identifiers from text before it is passed on, analysed or sent to a language model. Tools for this task exist. The best known open source one is Microsoft Presidio [2], which combines pattern recognisers with a statistical model for named entities.

Such tools are built around large languages and, for identifiers, largely around American formats. Luxembourg is a hard case for them. Administrative and private text is written in Luxembourgish, French, German and English, often within one document. Luxembourgish has few language resources, and no pipeline for it exists in the common libraries. The country has its own national identification number, the *matricule*, with two check digits, and its own phone and address conventions. Many residents and cross-border workers also carry French, Belgian or German identifiers.

A second problem is less often measured. A tool that marks too much is as unusable as one that marks too little, because a redacted text must remain readable. German and Luxembourgish write every noun with a capital letter, which removes the cheapest signal that a word is a name.

This paper asks two questions. First, can a system that puts rules first and uses a model only for what rules cannot know reach the coverage of a general tool on text in Luxembourg's formats, while marking less of the text that contains no personal data? Second, how far do results on a test set built by the developer carry over to names and documents the system has not seen? The second question arose during the work: an early version scored close to 1.00 on its own test set, and the paper reports what happened when the names in that test set were replaced.

The contributions are a description of the system, an evaluation on four kinds of data with a strict separation of development and test material, and a comparison with Presidio that states where each tool is ahead.

2 Materials and Methods

2.1 System

The system, called *tagging* [3], has a core written in Rust with bindings for Python and a build for WebAssembly. Each detector returns spans with a type, byte offsets and a score. Overlapping spans were resolved in favour of the longest and then the highest scoring one, and spans below a threshold of 0.5 were dropped.

Formats and check digits. Regular expressions proposed candidates for emails, URLs, IP addresses, phone numbers, IBANs, payment cards and dates, and each candidate was validated: IBANs by the ISO 7064 remainder, cards by the Luhn algorithm and issuer prefix, the Luxembourg matricule by its Luhn and Verhoeff digits. The French social security number, the Belgian national number and the German pension and tax numbers were validated by their own check digits. Codes shaped like passport numbers were accepted unless a word such as *order* or *invoice* preceded them. A date became a date of birth when a birth keyword stood nearby.

Word lists. First names (4,383), family names (19,516), places and organisations were taken from Wikidata [4]: the given and family names of people with one of nine citizenships, the localities of Luxembourg, sovereign states, well-known cities and organisations. Short hand-kept lists added Luxembourgish nicknames and local organisations.

Name grammar. A first name took the surname after it, a title marked the name after it, and a company form or a business word turned neighbouring capitalised words into an organisation. Family names never matched on their own.

Everyday words. Many names are also words (*Rose*, *Berg*, *Orange*). A list of about 190,000 everyday words was built from two sources. In 150 million characters of Wikipedia text per language, a word counted as everyday if it was mostly written in lowercase or, for German and Luxembourgish nouns, often followed a determiner. Because the Luxembourgish Wikipedia is small, all 73,900 word forms of the Lëtzebuerger Online Dictionnaire that are not proper nouns were added [5]. Names found in this list needed further evidence: a second name part, a title or a cue word such as *bei* or *in*.

Form fields and addresses. A reader recognised a known field name in one of the four languages, a separator and a value, as in `Reisepass: C4FH7K2P1` or `"password": "..."`. Address rules covered street and house number in French, German, Luxembourgish and English order, and several postcode formats.

Models. Two models for named entity recognition were optional. The accurate mode used GLiNER [6] in its multilingual variant for personal data, prompted with descriptive labels. Five additional labels such as *profession or role* were also requested and their spans discarded, so that a generic noun had a better label to compete for than *person*. The fast mode used the small pipelines of spaCy [7], with the language of each text guessed from its function words and the multilingual pipeline for Luxembourgish. A third mode accepted spans from either model. In all modes, model spans were post-processed: articles and titles were stripped, coordinated names were split, and a span made only of everyday words was dropped unless a word list knew a name in it.

2.2 Data

Four kinds of data were used (Table 1). The template set consists of sentences generated from hand-written templates and rendered in parallel in the four languages, with a development and an evaluation split from different random seeds. To test generalisation, the persons, organisations and places of the evaluation split were replaced by 40 first names, 40 surnames, 20 small firms and 20 towns written without consulting the word lists. Text without personal data came from the 80 entity-free sentences of the evaluation split and from 120 hand-written everyday sentences in which English and French nouns were capitalised as in a heading. WikiANN [8] in the split of Rahimi et al. [9] provided Wikipedia sentences with silver labels for persons, organisations and locations. The dataset *pii-masking-300k* [10] provided synthetic letters and forms with 27 labels.

2.3 Baseline

Presidio 2.2 was configured as a careful user would configure it: spaCy's large pipelines for English, French and German, the multilingual pipeline for Luxembourgish, organisations switched on, and phone regions extended to Luxembourg and its neighbours. Presidio was given the language of every text, whereas tagging guessed it. Because Presidio reports every date and time, tagging was run with the corresponding option for the comparison.

2.4 Measures

The main measure was coverage: the share of labelled items that at least one predicted span overlapped, whatever type the span carried. It answers the question a redaction poses, namely whether the item would have been removed. Over-redaction was measured as the number of predicted spans on text without personal data, with dates and times excluded for both tools. For the two external datasets, the share of predicted spans that touched a labelled item served as a rough precision; it is rough because both label sets are incomplete. Runtime was measured on two CPU cores with the models loaded.

3 Results

3.1 A saturated test set

On the original evaluation split, the rules alone covered 0.99 of all items and found 0.98 of the persons. After the names were replaced, the rules found 0.54 of the persons,

Table 1: Data used for testing. None of the test material was inspected during development.

Data	Test material	Development material
Template sentences	208 sentences with 504 items, names replaced	Development split (248 sentences)
Text without personal data	80 entity-free sentences of the evaluation split	120 everyday sentences
WikiANN	300 test sentences per language (en, fr, de, lb)	Validation split
pii-masking-300k	Documents 1 to 120 per language (en, fr, de)	Documents 2,001 to 2,150

Table 2: Recall of the correct type on the evaluation split before and after its names were replaced. Columns: rules alone, with the fast model, with GLiNER.

	Rules	+ fast	+ GLiNER
Person, original names	0.98	0.97	1.00
Person, unseen names	0.54	0.80	0.92
Organisation, unseen firms	0.06	0.35	0.81
Location, unseen towns	0.60	0.85	0.96
Formats and check digits	1.00	1.00	1.00
Person, text in lowercase	0.72	0.82	0.98

0.06 of the firms and 0.60 of the towns (Table 2). Only 48% of the new first names, 68% of the surnames and 5% of the firms were in a word list, against 82% to 100% of the original ones. Types with a format or a check digit were unaffected. The models carried most of the generalisation: GLiNER kept persons at 0.92 and firms at 0.81, the fast model at 0.80 and 0.35.

3.2 Comparison with Presidio

Table 3 and Figure 1 give the comparison. On the template sentences with unseen names, every mode with a model covered more than Presidio (0.93 to 0.97 against 0.87), and the difference was larger for the Luxembourgish sentences alone (0.92 to 0.97 against 0.82). Presidio found a quarter of the Luxembourg matricules; the rules found all of them. On the letters and forms, the rules alone covered 0.72 to 0.74 and the modes with a model 0.75 to 0.84, against 0.49 to 0.59 for Presidio.

On Wikipedia sentences the picture was mixed. With both models, tagging was ahead in English (0.86 against 0.79), level in French (0.82) and behind in German (0.85 against 0.87) and Luxembourgish (0.77 against 0.83). With a single model it was within two points of Presidio in English and French and clearly behind in German and Luxembourgish.

3.3 Over-redaction and runtime

On the 80 entity-free sentences Presidio marked 71 spans that were neither dates nor times, and tagging marked 10 to 16 depending on the mode. On the 120 everyday sentences the counts were 115 against 2 to 12. GLiNER on its own, without rules and filters, marked 92 and 151 spans on the same two sets. On Wikipedia sentences, 18% of the spans Presidio marked touched no labelled entity, against 3% to 8% for tagging.

The rules processed 1,000 words in about 4 ms. The fast mode took 0.09 s when texts were passed in one batch, GLiNER 7.8 s, and Presidio 0.43 s with one call per text.

4 Discussion

The first question can be answered with a qualified yes. With a model switched on, the system covered more than Presidio on text in Luxembourg's formats and on forms, and it marked far less of the text that held no personal data. The advantage has identifiable sources: check digits for local identifiers, a reader for labelled fields, and the list of everyday words. The last of these matters most for Luxembourgish, where the dictionary reduced the wrong hits on everyday sentences from nine to one.

The same filter explains where the system fell behind. An organisation whose name consists of dictionary words is dropped together with the plumbers and fire brigades, and in German and Luxembourgish, where every noun is capitalised, no rule based on capitals can tell the two apart. Presidio keeps both, which gives it higher coverage on Wikipedia sentences and many more wrong hits elsewhere. Which trade is better depends on the use: a leak is costly, but a text in which every third noun is blacked out will not be used.

The answer to the second question is a warning. A test set generated from frequent real names is largely contained in word lists built from Wikidata, even though the lists were not built from the test set. The rules scored 0.98 on persons and fell to 0.54 when the names changed. Reporting only the first number would have been wrong by a wide margin. Replacing the entities of an existing test set is a cheap check that is advisable for any system that relies on lists.

Limitations. The comparison is not equally fair in all rows. The rules for form fields and identifiers were developed on a different sample of the same dataset that served as a test, so they fit its format, although development and test coverage agreed (0.63 and 0.61 to 0.64 for the rules in the default configuration). WikiANN, in turn, resembles the training data of spaCy's pipelines, on which Presidio relies. Presidio can be extended with custom recognisers, and none were written here. The everyday sentences were written by the developer, and WikiANN's labels are produced automatically and contain errors. Most importantly, no real document was

Table 3: Presidio and four modes of tagging on held-out data. Coverage is the share of labelled items overlapped by a predicted span. Ranges are over languages.

	Presidio	Rules	+ fast	+ GLiNER	+ both
Coverage					
Template sentences, unseen names	0.87	0.68	0.93	0.95	**0.97**
The same, Luxembourgish only	0.82	0.68	0.92	**0.97**	**0.97**
Letters and forms (pii-masking-300k)	0.49 to 0.59	0.72 to 0.74	0.75 to 0.78	0.80 to 0.82	**0.80 to 0.84**
WikiANN, English	0.79	0.53	0.79	0.81	**0.86**
WikiANN, French	**0.82**	0.55	0.80	0.80	**0.82**
WikiANN, German	**0.87**	0.48	0.78	0.79	0.85
WikiANN, Luxembourgish	**0.83**	0.57	0.75	0.70	0.77
Spans marked in text without personal data					
80 entity-free sentences	71	**10**	13	13	16
120 everyday sentences	115	**2**	9	7	12
Share of predicted spans on a labelled item					
WikiANN	0.82	**0.97**	0.95	0.94	0.92
Letters and forms	0.77	**0.93**	0.85	0.89	0.83
Seconds per 1,000 words	0.43	**0.004**	0.09	7.8	7.7

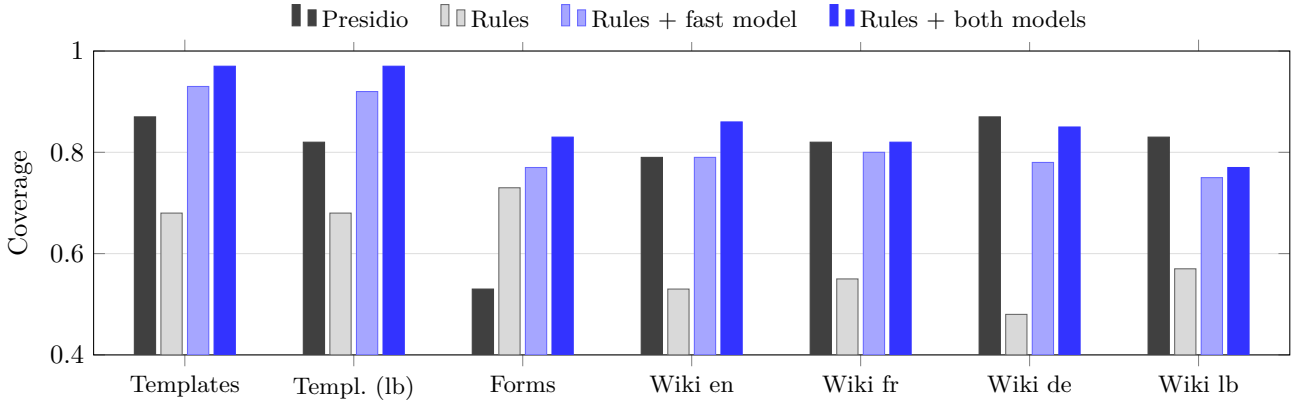


Figure 1: Coverage on held-out data. For the letters and forms the mean over English, French and German is shown.

used: every number comes from synthetic or encyclopedic text. At a recall of 0.92 per name, a document with ten names would still leak at least one of them in more than half of all cases, so the system should not be relied on for anonymisation without a check on the documents in question.

5 Conclusion

A system that validates what has a format, looks up what is listed and asks a model only for the rest can exceed a general tool on text in Luxembourg's languages and formats while marking a fraction of the text that contains no personal data. It does not exceed it everywhere: names made of everyday words in German and Luxembourgish remain the open problem, and the accurate model is slow on a CPU. The most useful methodological result is that a test set can be saturated by word lists without any deliberate leak, and that replacing its entities exposes this at little cost. The next step is an evaluation on real letters, emails and forms from an organisation in Luxembourg, labelled by hand, followed by a recogniser for Luxembourgish organisations that does not depend on capitals.

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Data and code availability

Code, word lists, the template data and all result files are available at `https://github.com/YannKleees/tagging`. The external datasets are downloaded by the evaluation scripts and are not redistributed.

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